

SIYANG LI

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EDUCATION

Johns Hopkins University

2020 – 2025

Doctor of Philosophy in Astronomy & Astrophysics

Master of Arts in Physics

2020 – 2023

Thesis Advisor: Professor Adam G. Riess

University of California, Berkeley

2016 – 2020

Bachelor of Arts, High Honors in Physics, cum laude

Thesis Advisor: Professor George F. Smoot III

FIRST-AUTHOR PUBLICATIONS

Total citations as of November 21, 2025: 178 ORCID: 0000-0002-8623-1082

12. **Siyang Li**, Adam G. Riess, Gagandeep S. Anand, Daniel Scolnic, Yukei S. Murakami, Dillon Brout, Erik R. Peterson. “The Complete Sample of Available SNe Ia Luminosity Calibrations from the TRGB Observed with either *HST* or *JWST*.” *ApJ*, 997, 115 (2026). <https://ui.adsabs.harvard.edu/abs/2025arXiv250408921L/abstract>
11. **Siyang Li**, Adam G. Riess, Daniel Scolnic, Stefano Casertano, Gagandeep S. Anand. “JAGB 2.0: Improved Constraints on the J-region Asymptotic Giant Branch-based Hubble Constant from an Expanded Sample of *JWST* Observations.” *ApJ*, 988, 97 (2025). <https://ui.adsabs.harvard.edu/abs/2025ApJ...988...97L/abstract>
10. **Siyang Li**, Gagandeep S. Anand, Adam G. Riess, Stefano Casertano, Wenlong Yuan, Louise Breuval, Lucas M. Macri, Daniel Scolnic, Rachael Beaton, Richard I. Anderson. “Tip of the Red Giant Branch Distances with *JWST*. II. I-band Measurements in a Sample of Hosts of 10 SN Ia Match *HST* Cepheids.” *ApJ*, 976, 177 (2024). <https://ui.adsabs.harvard.edu/abs/2024ApJ...976..177L/abstract>
9. **Siyang Li**, Adam G. Riess, Stefano Casertano, Gagandeep S. Anand, Daniel M. Scolnic, Wenlong Yuan, Louise Breuval, Caroline D. Huang. “Reconnaissance with *JWST* of the J-region Asymptotic Giant Branch in Distance Ladder Galaxies: From Irregular Luminosity Functions to Approximation of the Hubble Constant.” *ApJ*, 966, 20 (2024). <https://ui.adsabs.harvard.edu/abs/2024ApJ...966...20L/abstract>
8. **Siyang Li**, Adam G. Riess, Daniel Scolnic, Gagandeep S. Anand, Jiaxi Wu, Stefano Casertano, Wenlong Yuan, Rachael Beaton, Richard I. Anderson. “Standardized Luminosity of the Tip of the Red Giant Branch utilizing Multiple Fields in NGC 4258 and the CATs Algorithm.” *ApJ*, 956, 32 (2023). <https://ui.adsabs.harvard.edu/abs/2023ApJ...956...32L/abstract>
7. **Siyang Li**, Stefano Casertano, Adam G. Riess. “A *Gaia* Data Release 3 View on the Tip of the Red Giant Branch Luminosity.” *ApJ*, 950, 83 (2023). <https://ui.adsabs.harvard.edu/abs/2023ApJ...950...83L/abstract>
6. **Siyang Li**, Stefano Casertano, Adam G. Riess. “A Maximum Likelihood Calibration of the Tip of the Red Giant Branch Luminosity from High Latitude Field Giants using *Gaia* Early Data Release 3 Parallaxes.” *ApJ*, 939, 96 (2022). <https://ui.adsabs.harvard.edu/abs/2022ApJ...939...96L/abstract>

5. **Siyang Li**, Adam G. Riess, Michael Busch, Stefano Casertano, Lucas Macri, Wenlong Yuan. “A sub-2% Distance to M31 from Photometrically Homogeneous Near-Infrared Cepheid Period-Luminosity Relations Measured with the *Hubble Space Telescope*.” *ApJ*, 920, 84 (2021). <https://ui.adsabs.harvard.edu/abs/2021ApJ...920...84L/abstract>
4. **Siyang Li**, George F. Smoot III. “Characterization of a high efficiency silicon photomultiplier for millisecond to sub-microsecond astrophysical transient searches.” *SPIE Proc.* 11454, 1145422 (2020). <https://ui.adsabs.harvard.edu/abs/2020SPIE11454E..22L/abstract>
3. **Siyang Li**, George F. Smoot III, Bruce Grossan, Albert Wai Kit Lau, Marzhan Bekbalanova, Mehdi Shafiee, Thorsten Stezelberger. “Program objectives and specifications for the Ultra-Fast Astronomy observatory.” *SPIE Proc.* 11341, 113411Y (2019). <https://ui.adsabs.harvard.edu/abs/2019SPIE11341E..1YL/abstract>
2. **Siyang Li**, George F. Smoot III. “Characterization of a silicon photomultiplier for the Ultra-Fast Astronomy telescope.” *SPIE Proc.* 11115, 111150A (2019). <https://ui.adsabs.harvard.edu/abs/2019SPIE11115E..0AL/abstract>
1. **Siyang Li**, Jérôme Maire, Maren Cosens, Shelley A. Wright. “Detector characterization of a near-infrared discrete avalanche photodiode 5×5 array for astrophysical observations.” *SPIE Proc.* 11002, 110022G (2019). <https://ui.adsabs.harvard.edu/abs/2019SPIE11002E..2GL/abstract>

CO-AUTHOR PUBLICATIONS

Total citations as of November 21, 2025: 286

9. Stefano Casertano & the H_0 Distance Network Collaboration including **Siyang Li**. “The Local Distance Network: a community consensus report on the measurement of the Hubble constant at 1% precision.” *Submitted to A&A* (2025). <https://ui.adsabs.harvard.edu/abs/2025arXiv251023823H/abstract>
8. Adam G. Riess, **Siyang Li**, Gagandeep S. Anand, Wenlong Yuan, Louise Breuval, Stefano Casertano, Lucas M. Macri, Dan Scolnic, Yukei S. Murakami, Alexei V. Filippenko, Thomas G. Brink. “The Perfect Host: JWST Cepheid Observations in a Background-Free SN Ia Host Confirm No Bias in Hubble-Constant Measurements.” *ApJL*, 992, L34 (2025). <https://ui.adsabs.harvard.edu/abs/2025arXiv250901667R/abstract>
7. Rolf-Peter Kudritzki, Miguel A. Urbaneja, Fabio Bresolin, Lucas M. Macri, Wenlong Yuan, **Siyang Li**, Gagandeep S. Anand, Adam G. Riess. “The Hubble Constant Anchor Galaxy NGC 4258: Metallicity and Distance from Blue Supergiants.” *ApJ*, 977, 217 (2024). <https://ui.adsabs.harvard.edu/abs/2024ApJ...977..217K/abstract>
6. Adam G. Riess, Dan Scolnic, Gagandeep S. Anand, Louise Breuval, Stefano Casertano, Lucas M. Macri, **Siyang Li**, et al. “JWST Validates HST Distance Measurements: Selection of Supernova Subsample Explains Differences in JWST Estimates of Local H_0 .” *ApJ*, 977, 120 (2024). <https://ui.adsabs.harvard.edu/abs/2024ApJ...977..120R/abstract>
5. Gagandeep S. Anand, Adam G. Riess, Wenlong Yuan, Rachael Beaton, Stefano Casertano, **Siyang Li**, et al. “Tip of the Red Giant Branch Distances with JWST: An Absolute Calibration in NGC 4258 and First Applications to Type Ia Supernova Hosts.” *ApJ*, 966, 89 (2024). <https://ui.adsabs.harvard.edu/abs/2024ApJ...966...89A/abstract>
4. Daniel Scolnic, Adam G. Riess, Jiaxi Wu, **Siyang Li**, Gagandeep S. Anand, Rachael Beaton, Stefano Casertano, Richard I. Anderson, Suhail Dhawan, Xinwei Ke. “CATS: The Hubble Constant from Standardized TRGB and Type Ia Supernova Measurements.” *ApJL*, 954, L31 (2023). <https://ui.adsabs.harvard.edu/abs/2023ApJ...954L..31S/abstract>

3. Louise Breuval, Adam G. Riess, Lucas M. Macri, **Siyang Li**, Wenlong Yuan, Stefano Casertano, Tarini Konchady, Boris Trahin, Meredith J. Durbin, Benjamin F. Williams. “A 1.3% distance to M33 from *HST* Cepheid photometry.” *ApJ*, 951, 118 (2023). <https://ui.adsabs.harvard.edu/abs/2023ApJ...951..118B/abstract>
2. Jiaxi Wu, Daniel Scolnic, Adam G. Riess, Gagandeep S. Anand, Rachael Beaton, Stefano Casertano, **Siyang Li**. “Comparative Analysis of TRGBs (CATs) from Unsupervised, Multi-Halo-Field Measurements: Contrast is Key.” *ApJ*, 954, 87 (2022). <https://ui.adsabs.harvard.edu/abs/2023ApJ...954...87W/abstract>
1. Albert W. K. Lau, Mehdi Shafiee, George F. Smoot, Bruce Grossan, **Siyang Li**, Zhanat Maksut. “On-sky silicon photomultiplier detector performance measurements for millisecond to sub-microsecond optical source variability studies.” *JATIS*, 6(4), 046002 (2020). <https://ui.adsabs.harvard.edu/abs/2020JATIS...6d6002L/abstract>

OTHER PUBLICATIONS

Siyang Li, Rachael Beaton. “Tip of the Red Giant Branch.” Invited chapter in *The Hubble Constant Tension* (Eds. E. Di Valentino & D. Brout), Springer Singapore, 2024. DOI: 10.1007/978-981-99-0177-7_8.

CosmoVerse 2024 White Paper on Cosmic Tensions: *J-region Asymptotic Giant Branch*. Principal Author: **Siyang Li**; Co-authors: A. Riess, S. Casertano, G. Sloan, L. Breuval, B. Zgirski, G. Anand, R. Anderson, C. Huang, D. Scolnic. <https://ui.adsabs.harvard.edu/abs/2025arXiv250401669D/abstract>

CosmoVerse 2024 White Paper on Cosmic Tensions: *Tip of the Red Giant Branch*. Principal Author: R. I. Anderson; Co-authors: G. S. Anand, I. Saltas, G. de Somma, **S. Li**, V. Vansevicius, L. Breuval, A. Riess.

CosmoVerse 2024 White Paper on Cosmic Tensions: *Cepheids*. Principal Author: L. Breuval; Co-authors: R. I. Anderson, G. De Somma, L. Galbany, **S. Li**, L. Macri, L. Perivolaropoulos, A. Riess, V. Vansevicius.

TELESCOPE PROPOSALS

Refining a New Hubble Tension Probe: Testing J-region Asymptotic Giant Branch Homogeneity Assumptions

PI, *Lick Observatory Shane 3-meter Telescope* — 3 nights awarded

Building the Last Bridge Between the PHAT & PHATTER Surveys and Carbon Star Cosmology

PI, *Hubble Space Telescope Cycle 32 Bridge*, GO-17911 — 8 orbits awarded (\$25,000)

Towards Standardization of the J-region Asymptotic Giant Branch: Investigating Asymmetric Luminosity Functions with Carbon Star Spectra

PI, *Apache Point Observatory ARC 3.5-meter Telescope* — 7 half-nights awarded

A 1% cross-calibration of Cepheids, TRGB, and JAGB in five nearby galaxies with HST

Co-I (PI: L. Breuval), *Hubble Space Telescope Cycle 31*, GO-17520 — 33 orbits awarded

INVITED TALKS

Cross-Survey Cosmography: Precision Cosmology, Emerging Tensions and New Physics Conference, New Delhi, India Apr 2026

33rd Texas Symposium on Relativistic Astrophysics (memorial of G. F. Smoot III), Tempe, AZ Dec 2025

Brinson Prize Fellowship Workshop, Chicago, IL Nov 2025

<i>Tensions in Cosmology Conference, Corfu, Greece</i>	<i>Sep 2025</i>
<i>National Astronomical Observatory of Japan Cosmology Seminar, jointly with La Sapienza, Tokyo</i>	<i>Jul 2025</i>
<i>University of Tokyo Astronomy Department Colloquium, Tokyo, Japan</i>	<i>Jul 2025</i>
<i>Tohoku University Astronomy Department Colloquium, Sendai, Japan</i>	<i>Jun 2025</i>
<i>Sauverny Observatory (EPFL) Tea Time Talk, Versoix, Switzerland</i>	<i>May 2025</i>
<i>CosmoVerse Workshop@Naples, Naples, Italy</i>	<i>May 2025</i>
<i>ISSI Workshop: What's under the H0od? Towards Consensus on the Local Hubble Constant, Bern</i>	<i>Mar 2025</i>
<i>University of California, Davis Physics Department Seminar</i>	<i>Jan 2025</i>
<i>Yale University Cosmology Seminar, New Haven, CT</i>	<i>Nov 2024</i>
<i>CosmoVerse Seminar (https://youtu.be/V1P6jb4LiNE)</i>	<i>May 2024</i>
<i>Space Telescope Science Institute JWST Science Conversation, Baltimore, MD</i>	<i>Mar 2024</i>
<i>ISSI Annual Meeting: The Stellar Path to the H_0 Tension in the Gaia, TESS, LSST and JWST Era, Bern (virtual)</i>	<i>Jun 2023</i>

CONTRIBUTED PRESENTATIONS

<i>JASMINE Consortium Workshop: Dynamics and Evolution of the Galactic Center, Mitaka, Tokyo</i>	<i>Apr 2026</i>
<i>sys2025: Systematic and Measurement Errors across the Sciences, Huntsville, AL (oral)</i>	<i>Nov 2025</i>
<i>University of California, Berkeley Lunch Talk, Berkeley, CA (oral)</i>	<i>Oct 2025</i>
<i>JASMINE Consortium Meeting, Sendai, Japan (oral)</i>	<i>Aug 2025</i>
<i>Kyoto University Department of Astronomy Seminar, Kyoto, Japan (oral)</i>	<i>Jul 2025</i>
<i>Kavli IPMU APEC Seminar, Tokyo, Japan (oral)</i>	<i>Jul 2025</i>
<i>American Astronomical Society Meeting 245 Cosmology I, National Harbor, MD (oral)</i>	<i>Jan 2025</i>
<i>Cosmo'24, Kyoto, Japan (oral)</i>	<i>Oct 2024</i>
<i>Resolved Stellar Populations 2024 (Celebration of Peter Stetson's Career), Florence, Italy (oral)</i>	<i>Oct 2024</i>
<i>CosmoVerse@Kraków 2024, Kraków, Poland (oral)</i>	<i>Jul 2024</i>
<i>The 6th Neighborhood Workshop at Penn State University, State College, PA (oral)</i>	<i>Apr 2024</i>
<i>Johns Hopkins AstroCoffee, Baltimore, MD (oral)</i>	<i>Feb 2024</i>
<i>American Astronomical Society Meeting 243, New Orleans, LA (oral)</i>	<i>Jan 2024</i>
<i>Johns Hopkins Giant Branch Workshop, Baltimore, MD (oral)</i>	<i>Nov 2023</i>

<i>Statistical Challenges in Modern Astronomy VIII</i> , State College, PA (poster)	<i>Jun 2023</i>
<i>The 5th Neighborhood Workshop at Penn State University</i> , State College, PA (oral)	<i>Apr 2023</i>
<i>American Astronomical Society Meeting 241 Cosmology II</i> , Seattle, WA (oral)	<i>Jan 2023</i>
<i>iid2022 Workshop: Statistical Methods for Event Data</i> , Guntersville, AL (oral)	<i>Nov 2022</i>
<i>Johns Hopkins Physics and Astronomy Grad Wine & Cheese Talks</i> (oral)	<i>Nov 2022</i>
<i>American Astronomical Society Meeting 240 Cosmology III</i> , Pasadena, CA (oral)	<i>Jun 2022</i>
<i>Johns Hopkins Physics and Astronomy Grad Wine & Cheese Talks</i> (oral)	<i>Sep 2021</i>
<i>SPIE Astronomical Telescopes + Instrumentation, Conference 11454</i> (poster & oral)	<i>Dec 2020</i>
<i>UC Berkeley Summer Undergraduate Research Fellowship Conference</i> (oral)	<i>Aug 2019</i>
<i>SPIE Optics + Photonics, Conference 11115</i> , San Diego, CA (oral)	<i>Aug 2019</i>
<i>8th Applied Optics and Photonics China, Conference 10</i> , Beijing, China (poster)	<i>Jul 2019</i>
<i>Exploring the Energetic Universe Conference</i> , Nur-Sultan, Kazakhstan (oral)	<i>Jun 2019</i>
<i>SPIE Defense + Commercial Sensing, Conference 11002</i> , Baltimore, MD (poster)	<i>Apr 2019</i>
<i>UC Berkeley Physics Department Poster Session</i> (poster)	<i>Apr 2019</i>
<i>UC Berkeley Physics Department Poster Session</i> (poster)	<i>Apr 2018</i>

AWARDS AND FELLOWSHIPS

Brinson Prize Postdoctoral Fellowship (\$360,000)	<i>2025</i>
International Space Science Institute Postdoctoral Fellowship (declined)	<i>2025</i>
American Astronomical Society Rodger Doxsey Travel Prize <i>One of eight recipients selected from 47 entries; funded travel to AAS 245.</i>	<i>2024</i>
National Science Foundation Graduate Research Fellowship (\$108,000) <i>Five-year fellowship providing three years of full graduate school funding.</i>	<i>2022–25</i>
University of California, Berkeley Regents' and Chancellor's Scholarship <i>Awarded to the top 2% of undergraduates based on academic and extracurricular achievements.</i>	<i>2016–20</i>
SPIE Optics and Photonics Education Scholarship <i>Granted to outstanding individuals for potential long-term contributions to optics and photonics.</i>	<i>2019</i>
8th Applied Optics and Photonics China Best Poster Award	<i>2019</i>
Measurement Science Conference Scholarship	<i>2019</i>
University of California, Berkeley Summer Undergraduate Research Fellowship	<i>2019</i>
University of California, Berkeley Academic Opportunity Fund Grant	<i>2019</i>
University of California, Berkeley Regents' & Chancellor's Research Fellowship	<i>2017, 2018</i>

Hong Kong University of Science and Technology International Research Fellowship <i>Fully covers airfare, accommodation, and meals for research visits to HKUST.</i>	2017–19
University of California, Berkeley Physics Undergraduate Research Scholarship	2017–20

TEACHING ASSISTANTSHIPS

AS.171.118 Stars and the Universe: Cosmic Evolution , Johns Hopkins University	Spring 2021, 2022
AS.171.102 General Physics: Physical Science Majors II , Johns Hopkins University	Fall 2020
AS.173.111 General Physics Laboratory I , Johns Hopkins University	Fall 2020

SERVICE

<i>CosmoVerse Data Challenge & School Lead Organizer</i>	2025–26
<i>University of California, Berkeley Astronomy Department Spring Colloquium Committee</i>	2025
<i>Springer Nature Scientific Reports</i> , Referee	2025
<i>The Astrophysical Journal Supplement Series</i> , Referee	2024

OUTREACH

Astro Night Public Outreach Talk <i>Public talk for high school students, community members, and Sequoia Community College undergraduates.</i>	2025
Johns Hopkins Physics Fair Volunteer <i>Live physics demonstrations to make scientific concepts accessible to the public.</i>	2025
Johns Hopkins Physics Department Outreach Volunteer <i>Visited Towson High School to present physics demonstrations and discuss careers in astronomy.</i>	2025
Johns Hopkins Physics Department Mentorship Committee Member <i>Coordinates the mentorship program in the JHU Physics & Astronomy department.</i>	2024–25
Johns Hopkins Physics Department Mentor <i>Academic and career mentor for an undergraduate in the JHU Physics & Astronomy department.</i>	2023–24
PATHS Mentor <i>Study habits workshop and one-on-one guidance for an Alexandria City public school student.</i>	2021–22
ARCH Mentor and Panelist <i>Scholarship and essay writing panelist; one-on-one mentor for an underprivileged college applicant.</i>	2021

OTHER

Selected to attend the 73 rd Lindau Nobel Laureate Meeting (Physics), Lindau, Germany	Jun–Jul 2024
Siyang Li , George F. Smoot III, Thorsten Stezelberger. “Silicon Photomultiplier Telescope Camera.” U.S. Provisional Patent Ser. No. 62/940,250, filed November 26, 2019.	